The University of North Carolina at Chapel Hill

DATA 780: MACHINE LEARNING

Section: 973A

2026 Fall

Course Description

Course Description: Prerequisite, DATA 710, DATA 730; Pre- or corequisite, DATA 750. This course equips students with knowledge of existing tools for predictive analytics and foundational concepts in machine learning. The course covers core principles of machine learning and pattern analysis. Topics include maximum likelihood estimation, regression; classification; cross validation; generalization and overfitting; introduction to neural networks; nonparametric estimators; clustering; tree-based methods; autoencoders; kernel methods. Applications in tabular, image, and textual data for supervised and unsupervised learning tasks are covered.
Grading Status: Graduate Graded
Credit Hours: 3.00

Goals, Objectives, Learning Outcomes, Expectations

By the end of this course, learners will be able to:

1. Apply an ML framework to execute and adapt ML methods
2. Apply strategies and techniques to enhance model performance
3. Evaluate the strengths and limitations of ML methods
4. Create a formal ML approach (features, output, objective function, and cross-validation) to derive insights from real-world problems

Course goals include:

- Understand mathematical principles of machine learning methods
- Build machine learning models for both supervised and unsupervised learning tasks
- Evaluate machine learning models and understand model performance based on the domain and application

Grading (including grading scale, percentage breakdown of major assignments, and how attendance or participation will affect a student's final grade).

Assessment	Weight	Description
Participation	5%	Attendance and participation in class activities is considered.
Quizzes	5%	Quizzes evaluation basic understanding of contents from each unit
Homework Assignments	20%	3-4 assignments will be expected.
Group Project	25%	The final project is designed to give students the opportunity to explore machine learning methods and applications in a context that interests them.
Midterm	20%	Midterm examination
Final Exam	25%	Final examination (comprehensive exam)

Grading Scale

Final grades will not be rounded. Final course grades follow UNC Graduate School grading scale. MADS courses use the ranges described below:

Grade	Description	Numeric Value
H	High Pass: Clear excellence	93 – 100
P	Pass: Entirely satisfactory graduate work	80 – 92.9
L	Low Pass: Inadequate graduate work	70 – 79.9
F	Fail	0 – 69.9

Course Materials (physical and/or electronic) required for purchase

Not Applicable

Mandatory Disclaimer

This course engages diverse scholarly perspectives to develop critical thinking, analysis, and debate, and inclusion of a reading does not imply endorsement.